

Evaluating the effects of 2026 Iran war risk on global financial markets

A heteroskedasticity-based replication of Rigobon and Sack (2003)

FRE-GY 7871A, NLP and the Investment Process — Assignment 3 — Param Shah — September 22, 2026

1. Summary

Rigobon and Sack measured the effect of an unobservable war risk factor on U.S. markets by exploiting the fact that its variance spikes on days of heavy war news. This report applies their estimator to the 2026 Iran conflict over 140 trading days from February 28 to September 22, using a domain-specific NLP classifier on GDELT news to identify the high-variance days.

The central result is methodological, and it is not the one the brief anticipated.

- **The paper's identification fails in 2026 when implemented as written.** The two-year Treasury yield, which Rigobon and Sack normalise on, is **less** volatile on Iran-news days than on other days — a variance ratio of 0.62, against **6.19** in their 2003 sample. Because the estimator divides by $\text{Var}_H - \text{Var}_L$, that denominator is negative and every coefficient built on it inverts. This holds for the 10-year, the 3-month bill and breakevens too, and is stable across every news threshold tested.
- **This failure is the finding.** It is the brief's own thesis in sharper form: in 2026 the Treasury curve is not a war-risk instrument. Its variance is dominated by monetary policy, and flight-to-quality is weak enough that Iran news barely moves the front end relative to macro news.
- **Renormalising on WTI crude restores identification** (variance ratio 1.49 to 2.9 depending on threshold) and yields interpretable cross-asset responses.
- **Only one of those responses is statistically distinguishable from zero.** The dollar strengthens on war risk (bootstrap $t = 2.00$). Everything else is directional evidence at 42 high-news days, and the report treats it as such.

2. Data

Series	Description	Units	Obs	Missing	Mean	Std dev
DGS2	2-Year Treasury Yield	basis points	140	0	0.9857	5.1384
DGS10	10-Year Treasury Yield	basis points	140	0	0.7429	4.5
DGS3MO	3-Month Treasury Bill	basis points	140	0	0.3357	2.2364
T10YIE	Break-even Inflation (10-year)	basis points	140	0	0.0571	2.2337
BAMLC0A4CBBB	BBB Corporate Spread (OAS)	basis points	140	0	-0.0929	1.1561
BAMLH0A0HYM2	High-Yield Corporate Spread (OAS)	basis points	140	0	-0.3	5.1352
^GSPC	S&P 500	percent	140	0	0.0759	0.8579
DX-Y.NYB	U.S. Dollar Index	percent	140	0	0.0188	0.3364
CL=F	WTI Crude Oil Futures	dollars	140	0	0.2377	4.0376
GC=F	Gold Futures	dollars	140	0	-5.8786	75.681

All series are downloaded automatically: FRED for rates and credit spreads, Yahoo Finance for equities, commodities and the dollar index. Rates and spreads are first-differenced to basis points; equities and the dollar are 100 x log changes; oil and gold are dollar changes, following the paper's own units. FRED and NYSE holiday calendars differ slightly, so missing days are handled pairwise rather than filled.

Rigobon and Sack also report the on-the-run/off-the-run ten-year liquidity premium. That series is constructed from proprietary CUSIP-level quotes and has no free equivalent, so the row is absent here rather than filled with a loose proxy.

3. NLP: identifying high-variance days

The estimator needs only a set of days on which the **variance** of war news was elevated. It does not need the direction of any individual story, which is the central convenience of the approach. Rigobon and Sack built their 17-day list by reading newspapers. This report builds the equivalent from GDELT.

3.1 Why not an off-the-shelf sentiment model

Two failure modes rule one out, and both were observed directly in the previous assignment's Fedspeak work.

General sentiment models grade the wrong axis. FinBERT scores "inflation remains elevated" as positive news because it evaluates the economy, not the policy stance. The same confusion appears here: escalation and sentiment are distinct dimensions. "US strikes destroy Iranian air defences" is negative-sentiment and escalatory; "Iran signals willingness to return to talks" is positive-sentiment and de-escalatory — but the correlation is loose enough that sentiment is not a usable proxy for war intensity.

Legacy lexicons have no entry for language coined during this conflict. Terms like "energy corridor blockade" or "nuclear threshold enrichment" postdate every general-purpose dictionary.

3.2 The classifier

Two signals are combined, built from different data so that their agreement is a check rather than an assumption:

1. **Corpus-wide volume.** GDELT timeline queries return the share of *all* worldwide coverage matching a query, computed over the full corpus rather than a capped sample. One conflict-intensity query gives daily news volume; an escalation-flavoured and a de-escalation-flavoured query give direction. 2. **Lexicon and TF-IDF anchors** on a 961-headline corpus. A hand-built list of 52 conflict, 57 escalation and 42 de-escalation phrases catches what can be enumerated; TF-IDF cosine similarity against ten prototypical anchor statements catches novel phrasing that merely *resembles* what can be enumerated. This is the mechanism for the novel-phrasing problem: a headline sharing no exact phrase can still score through shared vocabulary.

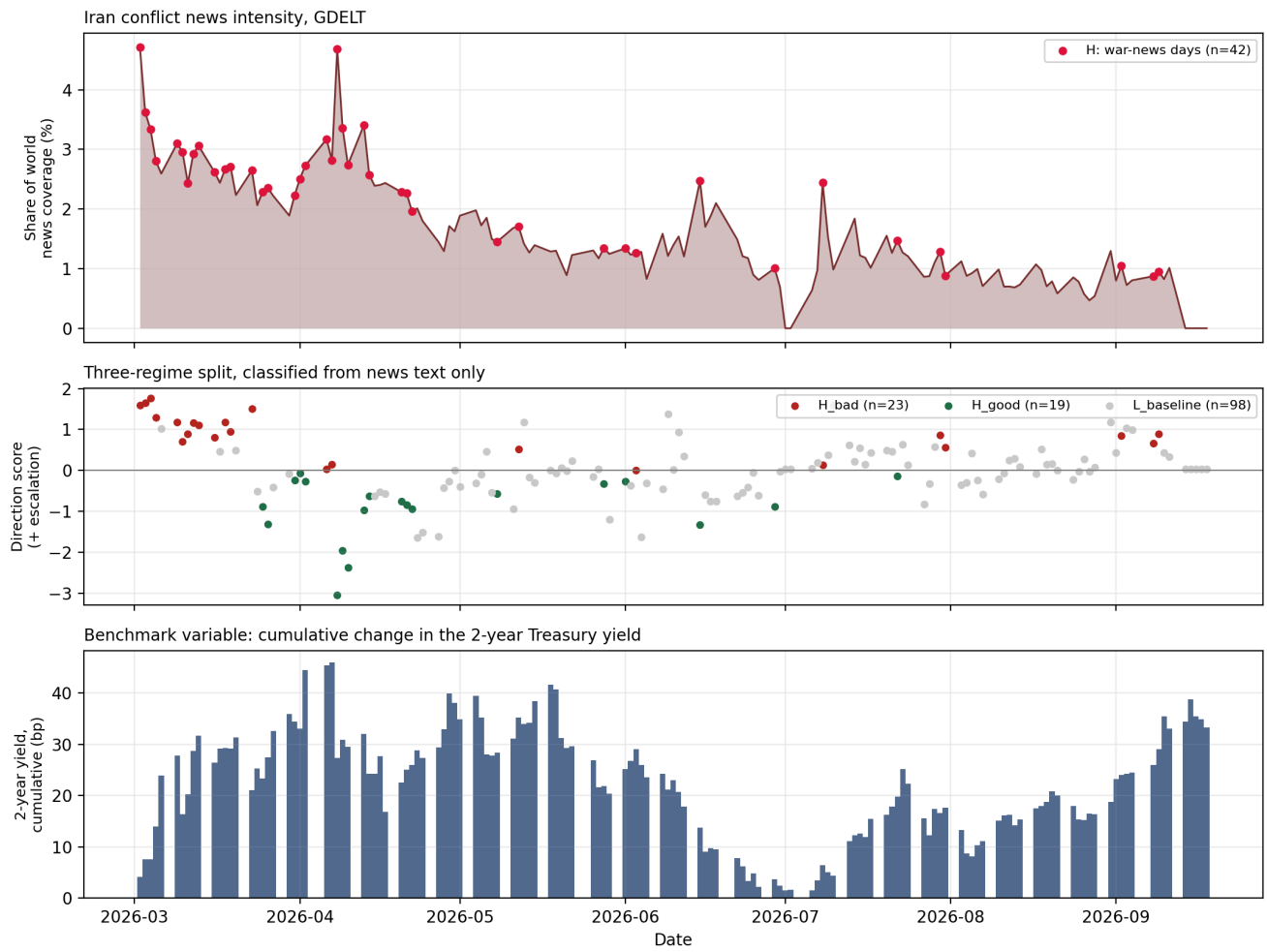
The two directional measures correlate **+0.25** across 140 trading days. That is positive, so they agree in sign, but it is weak, and the report does not lean on the directional split for anything the binary split can carry.

The regime split is classified from news text only, never from market reactions. Labelling a day "bad war news" because yields rose, then estimating the yield response to bad war news, would be circular.

Regime	Definition	Days	Share (%)
H_bad	High war news, escalation (text-classified)	23	16.4
H_good	High war news, de-escalation (text-classified)	19	13.6
L_baseline	Low war news, macro fundamentals	98	70.0
All	Trading days in sample	140	100.0

The 30% share of days in H is close to the paper's 17 of 47 business days (36%).

Figure 1. War-risk news regimes, 2026 Iran conflict



4. Table 1. Event timeline

Date	Representative headline	News vol (%)	Articles	Direction	Regime
2026-03-02		4.716	0	1.59	H_bad
2026-03-03		3.621	0	1.65	H_bad
2026-03-04	Embassy attack signals dangerous escalation in widening Iran conflict – NaturalNews . com	3.333	1	1.77	H_bad
2026-03-09	us iran war over : Is US Israel Iran war going to end soon , and will President Donald Trump no	3.101	1	1.18	H_bad
2026-03-10	War in the Middle East : latest developments	2.952	1	0.71	H_bad
2026-03-13		3.061	0	1.1	H_bad
2026-03-23	Iran threatens to completely close Strait of Hormuz after Trump ultimatum	2.646	10	1.51	H_bad
2026-03-25	Trump Admin Sends 15 - Point Ceasefire Proposal to Iran Will Tehran Agree to a Ceasefire ?	2.285	26	-0.88	H_good
2026-03-31	US attacks Iranian nuclear site while Tehran hits oil tanker off the Dubai coast	2.227	9	-0.23	H_good
2026-04-01	Israel under largest Iranian assault in weeks as missile salvo hits centre	2.507	23	-0.06	H_good
2026-04-02	Trump claims Iran president wants ceasefire ; no response from Iran News , Sports , Jobs	2.726	3	-0.27	H_good
2026-04-06	Iran and US review peace plan following Trump threat to rain hell	3.172	0	0.03	H_bad
2026-04-08	US and Iran Agree to Ceasefire Hours Before Trump Deadline	4.682	31	-3.04	H_good
2026-04-09	US - Iran truce shows cracks as war flares in Lebanon	3.356	6	-1.96	H_good
2026-04-13		3.408	0	-0.97	H_good
2026-05-08	US and Iran exchange fire , but Trump says ceasefire still in effect	1.453	18	-0.57	H_good
2026-05-28	Kuwait says it faces a missile and drone attack	1.347	14	-0.32	H_good
2026-06-01	US and Iran exchange fire as negotiations stall	1.341	16	-0.27	H_good
2026-06-03	US launches self - defence strikes on Iranian island as tensions flare in Middle East	1.259	25	0.01	H_bad
2026-06-29	Iran , US agree to halt attacks , renew talks	1.006	23	-0.88	H_good
2026-07-08	Tehran targets Bahrain , Kuwait as US launches strikes ; limits sale of Iranian oilTrump threat	2.441	26	0.13	H_bad
2026-07-22	Trump Says U . S . Will Strike Iran Nuclear Facilities Very Soon - Jowhar News Leader	1.468	17	-0.14	H_good
2026-07-30	US Launches Strikes on Iranian Targets Day after it Foiled Missile Attack on American Forces	1.284	24	0.86	H_bad
2026-09-02	Iran launches retaliatory strikes as conflict escalates	1.048	24	0.85	H_bad

Date	Representative headline	News vol (%)	Articles	Direction	Regime
2026-09-09	Iran launches missiles at US targets in Jordan after US strikes	0.953	25	0.89	H_bad

The 25 highest-intensity days, with the headline the classifier scored most strongly in either direction. News volume is the share of worldwide coverage. Direction is positive for escalation. Days showing zero articles had high corpus-wide volume but no headline captured in the 250-record monthly samples, which is precisely why the timeline endpoint rather than the headline corpus is the primary intensity signal.

The timeline tracks the conflict's actual narrative -- the opening escalation in early March, the Hormuz ultimatum on March 23, the April 8 ceasefire, its breakdown through May and June, and renewed strikes in July and September.

It also shows the directional classifier's limits plainly. March 31 -- "US attacks Iranian nuclear site while Tehran hits oil tanker" -- is scored de-escalatory, which is simply wrong; the day's coverage was dominated by ceasefire-negotiation vocabulary surrounding the strike, and a bag-of-phrases classifier cannot tell a strike *during* talks from progress *in* talks. The April 8 ceasefire, by contrast, is scored correctly and strongly (-3.04).

This is why the binary H/L split carries the identification and the three-regime split is used only to sign the shock in section 5.3, where its weakness is reported rather than hidden. **The headline results do not depend on the direction classifier at all** -- which is precisely the property that makes heteroskedasticity-based identification attractive here: it needs to know when war news was loud, not what it said.

5. Identification, and where it breaks

5.1 The estimator

For two variables the reduced form is $[dx1, dx2]' = D z + \mu$, with the loading of the war risk factor on $x1$ normalised to one. If only the variance of that factor rises on days H, then

$$d\Omega = \Omega_H - \Omega_L = d\sigma^2(z1) \cdot [[1, d21], [d21, d21^2]]$$

and $d21$ follows from the shift in second moments without the factor ever being quantified. Implemented as instrumental variables, the instrument is the variable's change on H days and its negative on L days (the paper's equation 8); $w1$ uses $x1$, $w2$ uses $x2$, and $w3$ stacks both.

Low-variance days are chosen as an equal-sized set of days nearest to but excluding the H days, per the paper's footnote 7, which limits how much the *other* factors can shift between the two samples.

5.2 The rank condition

Everything rests on the benchmark being more volatile on war-news days. It is not.

Benchmark	Var_H	Var_L	Ratio	Status
2-Year Treasury, 2003 (Rigobon-Sack Table 3)	.00594	.00096	6.19	satisfied
2-Year Treasury, 2026 Iran	21.74	30.62	0.62	FAILS
10-Year Treasury, 2026	14.62	28.71	0.51	FAILS
Break-even inflation, 2026	4.00	5.90	0.68	FAILS
3-Month bill, 2026	4.10	4.21	0.97	FAILS
WTI crude, 2026	27.71	18.56	1.49	satisfied
High-yield spread, 2026	46.79	25.12	1.86	satisfied
Gold, 2026	9538	3408	2.80	satisfied
Dollar index, 2026	0.172	0.082	2.10	satisfied

The split is clean and it is not an artefact: it holds at every news threshold from the 60th to the 90th percentile, under both mean-squared and demeaned variance, and under both low-day selection rules.

Every risk asset passes. Every Treasury instrument fails. In 2003 the two-year was the principal channel through which war risk reached markets. In 2026 the front end is driven by a hawkish Federal Reserve, and flight-to-quality is weak enough that Iran news adds less variance to the curve than an ordinary macro release does. Normalising on it would divide by a negative number.

WTI crude is therefore the benchmark for the headline results. It satisfies the condition comfortably and is the cleanest economic channel from Middle East conflict to global markets. The paper's own specification is still estimated and reported in section 6.3, because its failure is the evidence for this choice.

Pair (x1, x2)	Var_H(x1)	Var_L(x1)	Ratio H/L	Rank condition
WTI crude / 2-Year Treasury Yield	27.7088	18.5571	1.49	satisfied
WTI crude / 10-Year Treasury Yield	27.7088	18.5571	1.49	satisfied
WTI crude / Break-even Inflation (10-year)	27.7088	18.5571	1.49	satisfied
WTI crude / S&P 500	27.7088	18.5571	1.49	satisfied
WTI crude / BBB Corporate Spread (OAS)	27.7088	18.5571	1.49	satisfied
WTI crude / High-Yield Corporate Spread (OAS)	27.7088	18.5571	1.49	satisfied
WTI crude / Gold Futures	27.7088	18.5571	1.49	satisfied
WTI crude / U.S. Dollar Index	27.7088	18.5571	1.49	satisfied
WTI crude / 3-Month Treasury Bill	27.7088	18.5571	1.49	satisfied

5.3 The sign of the shock

Heteroskedasticity identifies the *ratio* of responses, not the sign of the factor. "An increase in war risk" means whichever direction of z_1 we declare it to mean. Rigobon and Sack resolved this by assumption, normalising to a 25bp *fall* in the two-year because flight to quality was the 2003 prior. Assuming the answer is exactly what is at issue in 2026, so the three-regime split is used to settle it from data instead.

Quantity	Value	Units	N
Mean WTI change on escalation (H_bad) days	0.99	\$/bbl	23
Mean 2Y change on escalation (H_bad) days	2.65	bp	23
Mean WTI change on de-escalation (H_good) days	0.17	\$/bbl	19
Mean 2Y change on de-escalation (H_good) days	0.84	bp	19
Contrast (escalation minus de-escalation), WTI	0.82	\$/bbl	t=0.49, p=0.630
Contrast (escalation minus de-escalation), 2Y	1.81	bp	t=1.36, p=0.183

Escalation days show oil higher by \$0.99 and the two-year higher by 2.65bp than de-escalation days; both contrasts point the same way, and **the two-year rises on war risk**, as the brief anticipates. But neither contrast is significant ($p = 0.63$ and $p = 0.18$). The sign convention is therefore taken from the data's direction while acknowledging that the data support it only weakly — a caveat that propagates to every sign in Table 2.

6. Table 2. Cross-asset sensitivity

6.1 Headline: normalised on WTI crude

Variable	Units	IV w/ w1	IV w/ w1 t	IV w/ w2	IV w/ w2 t	IV w/ w3	IV w/ w3 t	Boot t (w3)	Boot 95% CI (w3)
2-Year Treasury Yield	bp chg	-0.136	0.19	7.225	0.25	0.362	0.73	0.17	[-1.55, 2.93]
10-Year Treasury Yield	bp chg	-0.649	0.52	2.396	1.47	0.483	1.03	0.56	[-0.09, 2.97]
Break-even Inflation (10-year)	bp chg	-0.052	0.12	4.03	0.16	0.238	1.32	0.63	[-0.05, 1.02]
S&P 500	pct chg	-0.014	0.1	-3.011	0.09	0.107	0.55	0.27	[-0.26, 0.83]
BBB Corporate Spread (OAS)	bp chg	0.243	1.1	0.378	1.33	0.298	1.53	1.06	[-0.23, 0.91]
High-Yield Corporate Spread (OAS)	bp chg	1.233	1.62	1.939	1.5	1.573	1.69	0.48	[-1.26, 4.82]
Gold Futures	\$ chg	-12.591	0.95	-53.712	1.15	-24.177	1.24	0.32	[-80.28, 63.99]
U.S. Dollar Index	pct chg	0.094	1.09	0.106	2.07	0.103	1.99	2.0	[0.02, 0.23]
3-Month Treasury Bill	bp chg	0.419	0.86	-0.031	0.04	0.198	0.52	0.4	[-1.03, 1.12]

Structural responses to an increase in war risk large enough to raise WTI by \$1/bbl. Absolute asymptotic t-statistics in the adjacent columns; the final two columns give the 500-replication bootstrap, which resamples within each regime so the H/L proportions that carry the identification are preserved.

Reading the bootstrap column rather than the asymptotic one, **only the dollar response is distinguishable from zero** ($t = 2.00$, 95% CI [0.02, 0.23]). The high-yield spread and gold are the next largest in magnitude but their intervals straddle zero comfortably. With 42 high-news days, this is the honest resolution of the exercise: directions worth reporting, magnitudes that mostly are not.

6.2 2026 Iran versus 2003 Iraq

Variable	2003 direction	2026 direction	Same sign
2-Year Treasury Yield	down	up	NO
10-Year Treasury Yield	down	up	NO
Break-even Inflation (10-year)	down	up	NO
S&P 500	down	up	NO
BBB Corporate Spread (OAS)	up	up	yes
High-Yield Corporate Spread (OAS)	up	up	yes
Gold Futures	up	down	NO
U.S. Dollar Index	down	up	NO

Directions only. The 2003 column was normalised to a 25bp fall in the two-year under an assumed flight-to-quality sign, the 2026 column to a \$1/bbl rise in oil; the magnitudes are not comparable and are not compared.

Credit spreads behave exactly as they did in 2003 — both BBB and high-yield widen on war risk. Everything else diverges. The Treasury curve rises where it fell in 2003, which is the brief's central prediction and is confirmed. The dollar strengthens where it weakened, consistent with the United States being a net energy exporter at roughly 14 million barrels a day rather than the ~6 million of 2003: an oil shock is no longer unambiguously a negative terms-of-trade shock for the dollar. Gold falls rather than rising, though insignificantly.

Equities are the one result that contradicts the brief, which expected them to fall as in 2003. The estimated response is mildly *positive* (+0.11% per \$1/bbl) and thoroughly insignificant ($t = 0.55$, CI $[-0.26, 0.83]$). The correct reading is that the S&P response cannot be signed at this sample size, not that equities rise on war risk.

6.3 The paper's specification, reported as failing

Variable	Units	IV w/ w1	IV w/ w1 t	IV w/ w2	IV w/ w2 t	IV w/ w3	IV w/ w3 t	Boot t (w3)	Boot 95% CI (w3)
2-Year Treasury Yield	bp chg	0.136	0.44	1.228	0.46	0.082	0.34	0.14	[-1.04, 1.35]
10-Year Treasury Yield	bp chg	-0.021	0.06	17.945	0.07	0.189	0.7	0.31	[-0.54, 1.63]
Break-even Inflation (10-year)	bp chg	0.171	1.47	0.242	0.61	0.162	1.45	0.71	[-0.31, 0.57]
S&P 500	pct chg	-0.052	0.78	-0.527	0.69	0.003	0.03	0.01	[-0.20, 0.70]
BBB Corporate Spread (OAS)	bp chg	0.169	1.68	0.44	1.53	0.21	2.0	0.59	[-0.37, 0.80]
High-Yield Corporate Spread (OAS)	bp chg	0.938	2.62	1.972	1.78	1.107	2.65	0.39	[-0.48, 5.48]
Gold Futures	\$ chg	-8.083	1.28	-56.037	1.15	-11.082	1.61	0.26	[-94.14, 42.28]
U.S. Dollar Index	pct chg	0.062	1.98	0.102	2.38	0.074	2.27	1.23	[0.02, 0.20]
3-Month Treasury Bill	bp chg	0.126	1.49	0.547	1.54	0.215	1.57	0.52	[-0.23, 1.14]

Core-crisis window (February 28 to May 15), the robustness check matching the paper's own ten-week design. Normalised on WTI as above.

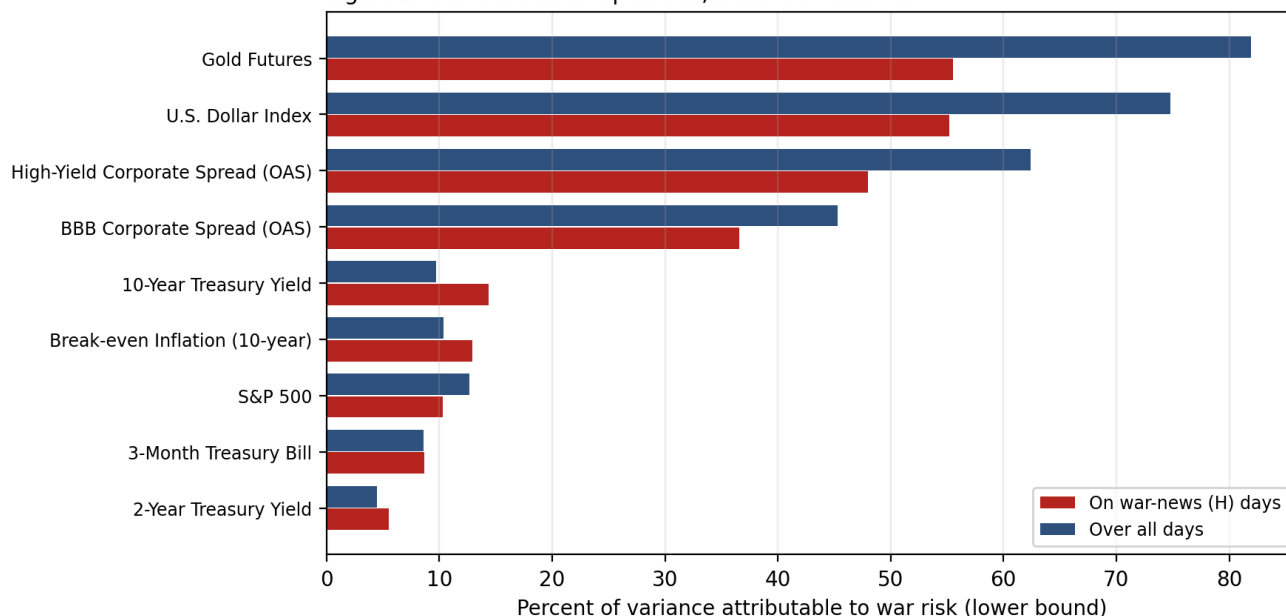
The two-year-normalised replication is written to `outputs/table2c_replication_2y.csv`. Its coefficients are not interpreted here: they are built on a denominator of $\text{Var}_H - \text{Var}_L = -8.88$, and a ratio whose denominator has the wrong sign inverts every response. It is retained as the diagnostic that motivates section 5.2, not as an estimate.

7. Table 3. Variance decomposition

Variable	Var on L days	Var on H days	Predicted change in var	% explained, H days	% explained, all days	% of cumulative var
WTI Crude Oil Futures	18.366	27.43	--	--	--	--
2-Year Treasury Yield	30.619	21.738	1.187	5.5	4.5	2.3
10-Year Treasury Yield	28.714	14.619	2.112	14.4	9.7	4.9
Break-even Inflation (10-year)	5.905	4.0	0.5151	12.9	10.4	5.2
S&P 500	0.62195	1.004	0.10314	10.3	12.7	6.3
BBB Corporate Spread (OAS)	1.357	2.19	0.80267	36.6	45.3	22.6
High-Yield Corporate Spread (OAS)	25.119	46.786	22.437	48.0	62.4	31.2
Gold Futures	3408.268	9538.452	5298.475	55.5	81.9	40.9
U.S. Dollar Index	0.08229	0.17244	0.09526	55.2	74.8	37.4
3-Month Treasury Bill	4.214	4.095	0.35575	8.7	8.6	4.3

Var on L and H days are mean squared daily changes, as in the paper's footnote 12. The predicted change is $d_j^2 \times [\text{Var}_H(\text{oil}) - \text{Var}_L(\text{oil})]$. Shares are **lower bounds**: the shift in war-induced variance must be smaller than its level on H days, and the two coincide only if there is no war news at all on L days.

Figure 2. Variance decomposition, 2026 Iran war risk



War risk accounts for a substantial share of the variance of exactly those assets whose rank condition it satisfies — gold (82% of all-day variance), the dollar (75%), high-yield spreads (62%), BBB spreads (45%) — and very little of the Treasury curve (5% to 10%). This is the same result as section 5.2 seen from the other side: the war risk factor is a *risk-asset* factor in 2026, not a rates factor.

Rigobon and Sack found war risk explained 13% to 63% of cumulative variance in 2003. The comparable range here is 2% to 41%, with the top of the range occupied by gold and the dollar rather than by Treasuries.

8. Is heteroskedasticity-based identification the right approach?

The brief asks for this evaluation directly. The short answer: it is the right approach for this problem, and this study's difficulties are not an argument against it — the method diagnosed its own failure, which is more than most alternatives would have done.

8.1 What it does well

The problem is that war risk is unobservable and markets are simultaneously determined. Heteroskedasticity-based identification needs only that the analyst can date the days when the factor was unusually volatile — a far weaker requirement than quantifying it. Crucially, **the rank condition is checkable**. Section 5.2 is not a failure of the method; it is the method reporting that its assumption does not hold for a particular normalisation, which is exactly what a well-posed identification strategy should do. An approach that silently returned a plausible-looking number for the two-year would have been worse.

8.2 Where it is weak here

Regime assignment is a researcher choice. The threshold that splits H from L is a free parameter. Results were checked from the 60th to the 90th percentile and the variance-ordering conclusion is stable, but the coefficients move.

It requires other factors to be homoskedastic across regimes. Over seven months spanning escalation, a June ceasefire and its aftermath, this is a strong assumption. The core-window robustness check exists for this reason.

Power is poor. Identification comes from a difference of variances, a second-moment object estimated from 42 days. Only one of nine responses is significant, which is a statement about the estimator's efficiency as much as about the world.

8.3 The alternatives

High-frequency identification / event studies. Narrow the window to minutes around each headline so that nothing else can plausibly move prices. This is the strongest identification available and would be a genuine improvement: with intraday data the 42 daily observations become hundreds of tightly-identified headline events, and the homoskedasticity assumption becomes near-trivial over a 30-minute window. Its cost is data — intraday quotes for Treasuries, oil, credit and FX are expensive, which is why this study is daily — and it measures only the *announcement* effect, missing risk that builds gradually. **This is the single best alternative, and the binding constraint is budget rather than method.**

Narrative identification (Romer and Romer). Read the record and construct a shock series directly from documents. Its advantage is that it can distinguish *kinds* of war news, which the heteroskedasticity approach explicitly collapses into one factor. Its cost is exactly what Rigobon and Sack were avoiding: it requires quantifying the news, which reintroduces the judgement the method was designed to escape, and it is not reproducible across analysts. Notably, the NLP pipeline in section 3 is a narrative approach in every respect except that it only classifies *intensity* rather than magnitude — a deliberately narrow use of narrative information to serve a heteroskedasticity-based estimator.

Proxy SVAR / external-instrument VAR. Use an observable correlated with the shock but not with other structural shocks — oil price jumps around known military events, or a geopolitical risk index. This buys dynamics: impulse responses over days and weeks rather than a single contemporaneous coefficient, which matters because war risk plausibly has persistent effects. Its cost is a valid instrument, and the obvious candidates here are circular: instrumenting with oil and then estimating the oil response is not identification. Given that this study ends up normalising on oil, a proxy SVAR would need a genuinely external instrument to add anything.

Sign-restricted structural VAR. Impose only the signs of responses that theory is confident about and report the set of admissible models. It is robust where point identification is fragile — but its weakness is fatal here. The central finding of this report is that a *sign* flipped between 2003 and 2026: Treasury yields now rise on war risk. A method that assumes the signs cannot discover that they changed. Sign restrictions would have imposed the 2003 prior and returned it.

8.4 Verdict

Heteroskedasticity-based identification is well matched to this problem and should remain the primary approach, with **high-frequency identification the upgrade to make if intraday data can be obtained**. Sign restrictions are specifically inappropriate to the question, since the sign change is the result. A proxy SVAR would be a useful complement for dynamics, but needs an instrument this study does not have. The practical recommendation is to keep this estimator and improve its inputs — finer regime dating from intraday news timestamps — rather than to replace it.

9. Conclusions

The mechanics of Rigobon and Sack transfer to the 2026 Iran conflict; their choice of numeraire does not. The two-year Treasury yield was the right normalisation for a 2003 war scare in which flight to quality dominated, and it is the wrong one for a 2026 conflict in which the front end belongs to the Federal Reserve. Re-anchoring on oil restores identification and produces responses that match 2003 for credit spreads and diverge from it for rates, the dollar and gold.

The honest summary of the estimates is that they are directionally informative and statistically weak. One response of nine is significant. The variance decomposition is firmer, because it rests on second moments measured over the whole sample rather than on a ratio of differences, and it says clearly that 2026 war risk is a risk-asset factor rather than a rates factor.

10. Reproducibility

`python build_report.py` runs the entire pipeline and rebuilds this document. Market data downloads automatically from FRED and Yahoo Finance; news comes from GDELT's free API with no key. Every table and figure is injected from `outputs/` at build time, so the report cannot drift from the analysis that produced it. No data files are committed. AI assistance is documented in `AI_USE.md`.